Highlights

UAV-TAlign-12K: Benchmarking Scene-Level Reliability in Cross-FOV UAV Infrared-Visible Alignment

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- UAV-TAlign-12K offers 6,039 synchronized cross-FOV RGB-infrared pairs.
- Seven baselines reveal broadly available pairwise homography evidence.
- UAV-TAlign yields 15 scene transforms and nine high-confidence products.
- Resampling and perturbations probe consensus stability and QA sensitivity.

UAV-TAlign-12K: Benchmarking Scene-Level Reliability in Cross-FOV UAV Infrared-Visible Alignment

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Abstract

Cross-field-of-view infrared-visible alignment is a core capability for UAV inspection, nighttime sensing, and multimodal scene understanding. We introduce UAV-TAlign-12K, a benchmark-scale collection of 6,039 synchronized pairs of RGB and thermal infrared images acquired across 15 industrial and natural scenes, together with an integrity-checked 6,037-pair evaluation split. The benchmark organizes evaluation at three complementary levels: pairwise homography evidence, scene-scale consensus, and reliability–coverage operating analysis. Seven representative methods span classical keypoint pipelines, infrared-visible-specific matching, and modern dense or learned matchers. Pairwise evidence is broadly available: RIFT2, RoMa, and raw MINIMA produce homographies for all 6,037 evaluated pairs, while LoFTR and XoFTR exceed 99% availability. Building on this evidence, UAV-TAlign combines progressive evidence scheduling, geometry-informed frame weighting, robust scene consensus, and cross-modal quality verification. It estimates transforms for all 15 scenes and identifies nine high-confidence alignment products at the canonical operating point, covering 74.4% of the evaluated pairs. Five-fold leave-frame-out analysis yields a median scene-wise mean displacement of 1.39 pixels, while controlled translation, rotation, and scale perturbations progressively reduce the mean joint quality signal. UAV-TAlign-12K establishes a reproducible foundation for benchmark-scale, confidence-aware UAV infrared-visible alignment.

Keywords: infrared-visible dataset, UAV imagery, multimodal registration, alignment benchmark, scene-level reliability

1. Introduction

Infrared-visible alignment is a foundational capability for UAV inspection, nighttime monitoring, and multimodal thermal sensing. Visible imagery provides semantic and structural detail, while thermal infrared imagery provides complementary information under changing illumination and low-light conditions. This combination supports multispectral detection, tracking, and aerial perception across demanding operating environments [1, 2, 3, 4, 5, 6, 7, 8]. Accurate cross-sensor geometry is therefore essential for treating the two streams as one sensing system rather than independent observations.

UAV acquisition introduces a distinct alignment problem. Cross-FOV optics, unequal sensor resolutions, changing viewpoints, and zoom variation reshape the same scene differently in the visible and infrared channels. Time-synchronized capture preserves these cross-sensor differences rather than geometrically registering the images. Infrared rendering and day-to-night appearance changes further alter local texture and contrast. A flight sequence consequently provides heterogeneous geometric evidence: individual frame pairs may differ substantially in spatial support, homography stability, and cross-modal agreement. Effective alignment must consolidate this evidence into a scene transform that can support downstream sensing.

Modern correspondence estimation has substantially increased the availability of pairwise evidence. Transformer, dense, and multimodal matchers such as LoFTR, LightGlue, DKM, RoMa, XoFTR, and MINIMA recover correspondences across increasingly difficult appearance changes [9, 10, 11, 12, 13, 14, 15, 16]. Recent UAV alignment benchmarks have also introduced synthetic transform supervision and direct geometric evaluation [17]. These advances establish a strong pairwise foundation. They also motivate a broader benchmark question: how should variable-quality frame estimates be consolidated and assessed at the scale of a captured UAV scene?

We address this question with UAV-TAlign-12K, a real captured cross-FOV UAV infrared-visible benchmark comprising 6,039 synchronized pairs across 15 industrial and natural scenes. Its official evaluation split contains 6,037 integrity-checked pairs spanning day, night, and low-light capture; grayscale and pseudocolor infrared rendering; and wide, zoom, and standard UAV views. The benchmark defines three complementary tracks. Pairwise evaluation measures homography evidence availability, scene-level evaluation measures multi-frame consensus products, and a reliability–coverage profile

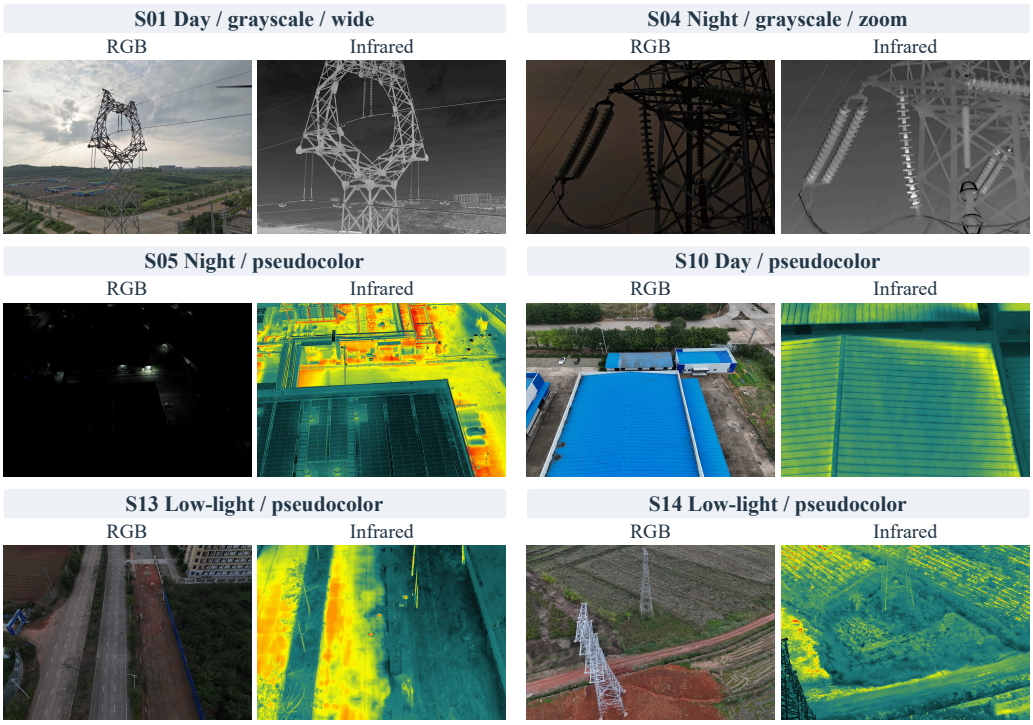

Figure 1: UAV-TAlign-12K at a glance. Representative RGB-infrared pairs from the same M350/H30T platform span day, night, and low-light capture, grayscale and pseudocolor rendering, and wide and zoom views. The collection contains 6,039 synchronized pairs; 6,037 form the integrity-checked official evaluation split.

characterizes selectable operating regimes. This hierarchy connects correspondence generation to the scene products required by UAV sensing systems.

Figure 1 presents representative synchronized image pairs and the conditions represented in the official evaluation split.

UAV-TAlign serves as the benchmark reference framework. It operates above a strong pairwise matcher and combines progressive evidence scheduling, geometry-informed frame weighting, robust scene consensus, and cross-modal quality verification. Across the official split, infrared-specific and modern matchers provide highly available pairwise homographies. UAV-TAlign converts this evidence into transforms for all 15 scenes and identifies nine high-confidence scene products at the canonical operating point, covering 74.4% of the evaluated pairs. Controlled perturbations and leave-frame-out resampling characterize quality-signal sensitivity and consensus stability.

Our contributions are threefold:

- We introduce UAV-TAlign-12K, a captured benchmark with 6,039 synchronized UAV RGB-infrared pairs across 15 scenes and an integrity-checked 6,037-pair evaluation split.
- We formulate a multi-level protocol spanning pairwise homography evidence, scene-scale consensus, and reliability–coverage analysis, supported by condition-aware diagnostics, resampling stability, and controlled perturbations.
- We present UAV-TAlign, a reference framework that converts strong pairwise evidence into high-confidence scene products through progressive evidence scheduling, robust consensus, and cross-modal quality verification.

2. Related Work

2.1. Cross-Modal Correspondence and Geometric Estimation

Classical registration combines local feature extraction, descriptor matching, and robust model estimation. Scale-space, binary, and nonlinear diffusion features provide efficient geometric references across diverse imaging conditions [18, 19, 20, 21, 22, 23]. RANSAC and its accuracy-oriented successors remain central to converting noisy correspondences into geometric models [24, 25, 26, 27]. For infrared-visible imagery, radiation-variation-insensitive constructions such as RIFT improve descriptor repeatability across modality-specific appearance changes [28]. RIFT2 reduces its computational cost through a revised rotation-invariant construction [29]. These methods provide an interpretable foundation for separating correspondence quality from downstream scene aggregation.

Learned correspondence methods have expanded both the robustness and spatial density of pairwise evidence. Early trainable detectors and descriptors established data-driven local representations [30, 31, 32, 33, 34, 35, 36, 37, 38]. Graph reasoning, transformers, and dense warp prediction subsequently enabled direct matching across low texture, repeated structure, and large appearance changes [9, 39, 40, 41, 10, 11, 12, 13, 14, 42, 43]. XoFTR and MINIMA further target visible-thermal or broadly multimodal correspondence [15, 16]. UAV-TAlign builds on this progress by treating pairwise matchers as evidence generators within a scene-scale geometric pipeline.

2.2. Infrared-Visible Alignment

Infrared-visible registration spans feature-based matching and learned correspondence estimation [28, 29, 15, 16]. More broadly, learned image alignment and stitching estimate transforms or reconstruct aligned imagery from training data [44, 45]. UAV dual-sensor imagery adds cross-resolution optics, viewpoint displacement, infrared rendering variation, and illumination changes. Registration under these conditions has motivated dedicated pairwise methods and UAV-specific data resources.

UAVMatch provides synthetic transform supervision and controlled geometric metrics for trainable UAV alignment [17]. ATR-UMMIM contributes 7,969 real UAV visible–infrared–registered-visible triplets with condition attributes and a semi-automated pixel-level reference [46]. UAV-TIRVis provides 80 manually registered aerial thermal-visible pairs for image-domain evaluation [47]. UAV-TAlign-12K complements these resources with a 6,039-pair captured cross-FOV collection and a multi-frame protocol that connects pairwise homographies, scene consensus, and reliability–coverage operating analysis.

2.3. UAV Multimodal Benchmarks and Scene-Scale Evaluation

Multimodal datasets such as KAIST, FLIR, LLVIP, M3FD, DroneVehicle, VTUAV, LasHeR, and Anti-UAV have advanced detection, fusion, tracking, and low-light perception [1, 2, 3, 4, 5, 6, 7, 8]. Their success demonstrates the value of synchronized visible and infrared sensing. Alignment, however, requires an explicit geometric product and benefits from evaluation beyond isolated correspondence counts.

UAV-TAlign-12K defines pairwise, scene, and operating-profile tracks. The pairwise track measures the availability and support of homography evidence. The scene track measures how multiple frame estimates form a consensus transform. The operating profile then orders scene products using fixed, interpretable cross-modal diagnostics. This design places classical, infrared-specific, and modern learned evidence generators within a common captured-data protocol, with UAV-TAlign as the scene-level reference framework [48, 49].

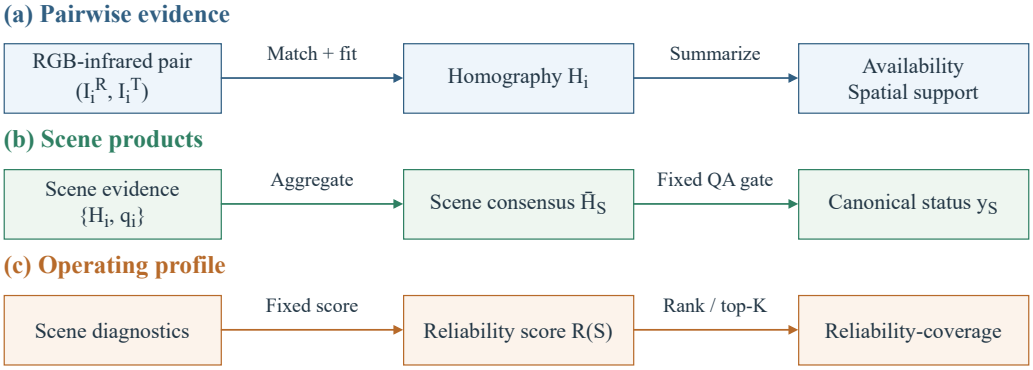

Figure 2: Three complementary benchmark tracks. Pairwise homography evidence is evaluated before multi-frame aggregation; the fixed canonical gate selects scene products; an independent score orders scenes for coverage-aware analysis. All tracks use the same official manifest. Per-scene counts and acquisition attributes are detailed in Appendix B. The editable schematic was prepared with OpenAI Codex assistance.

3. UAV-TAlign-12K and Evaluation Protocol

3.1. Dataset Scope

UAV-TAlign-12K is a benchmark for real captured UAV infrared-visible alignment. The collection contains 6,039 synchronized RGB-infrared pairs, corresponding to 12,078 images across 15 scenes. Its official evaluation split contains 6,037 integrity-checked pairs and 12,074 images. The collection spans cross-FOV and cross-resolution sensing under day, night, and low-light capture, grayscale and pseudocolor infrared rendering, and wide, zoom, and standard UAV views.

All journal-scale experiments use the same versioned manifest. UAV-TAlign-1K-Lite is retained as a fixed development and ablation subset, while UAV-TAlign-12K defines the benchmark-scale evaluation in this study. The official split comprises 7 day, 5 night, and 3 low-light scenes, together with 8 grayscale and 7 pseudocolor infrared scenes. Paired wide and zoom views provide controlled cross-FOV comparisons within the substation scene family.

3.2. Acquisition and Curation

UAV-TAlign-12K was acquired between the second half of 2024 and the first half of 2026 across industrial and natural scenes in southern China. All scenes were captured with the same DJI Matrice 350 RTK platform and Zenmuse H30T multisensor payload. Its uncooled VOx microbolometer operates in the 8–14 μm long-wave infrared (LWIR) band [50]. Throughout this

Table 1: Positioning of UAV-TAlign-12K relative to representative infrared-visible and UAV multimodal resources. UAVMatch denotes the synthetic transform-supervised UAV alignment benchmark.

Resource	Data and alignment reference	Scale	Evaluation focus
KAIST / FLIR / LLVIP / M3FD	Ground or mixed RGB-infrared data; paired or aligned imagery	Dataset-specific	Detection, fusion, low-light perception
DroneVehicle / VTUAV / LasHeR / Anti-UAV	UAV or tracking-centric RGB-infrared data; task annotations	Dataset-specific	Multimodal detection or tracking
UAVMatch [17]	Synthetic affine / homography supervision from UAV data	81K train / 15K test pairs	Transform regression and geometric metrics
ATR-UMMIM [46]	Real UAV triplets with registered-visible images; semi-automated pixel reference	7,969 triplets	Pairwise registration and condition attributes
UAV-TIRVis [47]	Real aerial thermal-visible pairs; manual registration	80 pairs	Image-domain registration quality
UAV-TAlign-12K	Captured cross-FOV UAV pairs; multi-frame scene protocol	6,039 candidate / 6,037 evaluated pairs	Pairwise evidence, scene consensus, operating coverage

benchmark, infrared images are the grayscale or pseudocolor renderings from this thermal channel. The integrated payload automatically synchronized the visible and infrared capture in time. Images retain their native cross-sensor geometry without manual geometric alignment. The flights combined hovering and route-based capture using a registered aircraft, a CAAC-qualified pilot, and permitted flight areas.

The released streams retain the native H30T acquisition modes. Wide-camera stills are 4032×3024, zoom-camera stills are 3664×2748, and infrared images are 1280×1024. In scenes 01 and 02, 3840×2160 still images were triggered while H30T video recording was active. Scene 07 is a synchronized video-derived sequence sampled at 2 frames per second over one complete

orbit around a transmission tower. Grayscale scenes use the H30T *White Hot* setting, and pseudocolor scenes use the on-device *Hot Iron* palette.

Curation criteria were fixed before algorithm evaluation and covered scene relevance, exposure, focus, and file integrity. The resulting candidate collection contains 6,039 synchronized pairs; the integrity audit defines the 6,037-pair official evaluation split. Selection did not use matcher outputs, alignment estimates, or quality scores.

Release preparation removes embedded metadata and normalizes filenames while preserving image pixels. RGB and infrared files are linked through their shared H30T capture key, ordered by capture time and sequence index, and assigned a common six-digit pair identifier. The release applies no resizing, cropping, rotation, recompression, or image enhancement. Nanning Normal University and Guangxi University jointly own the collection, which is prepared for release under the Creative Commons Attribution 4.0 International license.

3.3. Dataset Organization

Each scene has dedicated `rgb/` and `thermal/` directories, and paired samples share the same filename stem. The official manifest records scene identity, pair identity, illumination, infrared rendering, view type, and scene label. A scene groups frames sharing a site or target context and acquisition condition. It may represent a continuous sequence or multiple sorties under the same scene definition. Scene 07 is the documented single-orbit video sequence.

One manifest supports all pairwise, scene, and condition analyses. It also maps every reported result to its source scene and pair identifier.

3.4. Multi-Level Evaluation Protocol

The benchmark defines three complementary tracks. Track A evaluates each of the 6,037 pairs independently and measures the availability and support of pairwise homography evidence. Track B aggregates frame estimates into a scene transform and reports the canonical set of high-confidence scene products. Track C orders scenes with a fixed, interpretable reliability score to construct coverage-aware operating profiles.

Together, the tracks connect matcher output to deployable scene geometry. Pairwise measurements characterize evidence generation, scene-level measurements characterize multi-frame consensus, and the operating profile exposes how product coverage changes with scene quality.

3.5. Label-Free Scene-Consistency Diagnostics

The protocol measures cross-modal agreement after applying the scene consensus and compares it with a size-matched infrared reference. This reference resizes the infrared image to the RGB evaluation grid without estimating alignment. Diagnostics include edge-overlap improvement, gradient-NCC improvement, the fraction of frames with improved gradient agreement, severe baseline-relative outlier count and ratio, accepted-frame support, homography dispersion, and robust-rejection ratio. Appendix A specifies the reference, probe frames, normalization, and aggregation conventions used throughout the evaluation.

These signals jointly characterize cross-modal improvement, geometric support, and consensus stability. The resulting diagnostics support canonical product selection, condition-aware analysis, and reliability–coverage visualization under a common evaluation interface [48, 49].

4. UAV-TAlign Reference Framework

4.1. Problem Formulation

UAV-TAlign instantiates the scene-level track of UAV-TAlign-12K. For each scene, the input is a set of paired RGB and infrared UAV frames $S = \{(I_i^R, I_i^T)\}_{i=1}^N$, with superscripts R and T denoting RGB and thermal infrared, respectively. Viewpoint variation, cross-resolution sensing, infrared appearance changes, and illumination variation produce frame estimates with heterogeneous support. The goal is to recover an infrared-to-RGB scene homography and characterize its operating confidence.

This formulation casts registration as multi-frame evidence consolidation rather than single-pair selection. The framework converts uneven frame estimates into a geometrically stable and cross-modally consistent scene transform. For a selected frame pair, the infrared-to-RGB evidence is represented as a homography $H_i \in \mathbb{R}^{3\times3}$ acting on homogeneous image coordinates,

$$\tilde{\mathbf{p}}_{ij}^R \sim H_i \tilde{\mathbf{p}}_{ij}^T, \qquad H_i = \arg\min_H \sum_{j\in\mathcal{I}_i} \rho\Big(\big\| \pi(H\tilde{\mathbf{p}}_{ij}^T) - \mathbf{p}_{ij}^R \big\|_2^2 \Big), \tag{1}$$

where $\mathcal{I}_i$ denotes the inlier set selected by robust fitting, $\pi(\cdot)$ converts a homogeneous coordinate to image coordinates, and $\rho(\cdot)$ is the robust estimator loss. The scene-level problem is then to estimate a consensus transform $\bar{H}_S$ from the evidence set $\{H_i\}$ and assign its canonical product status.

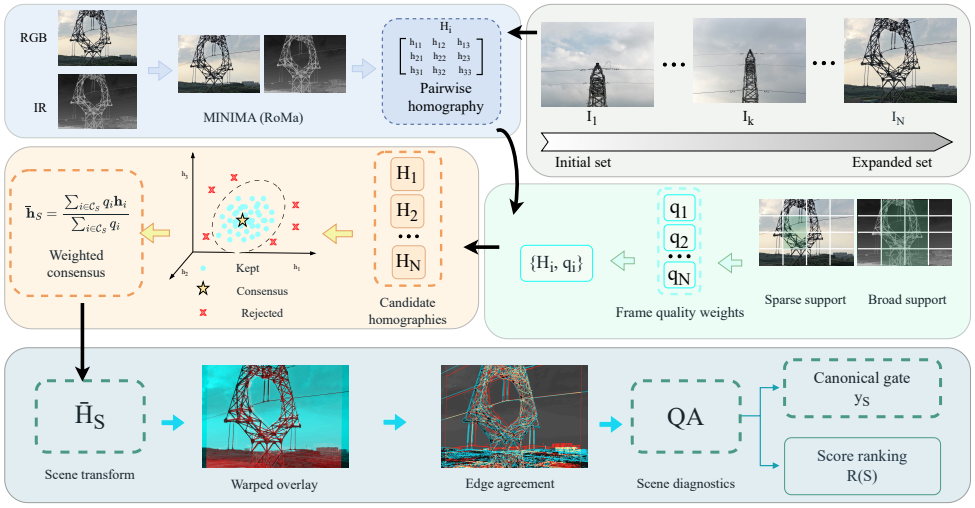

Figure 3: UAV-TAlign workflow. Progressive evidence selection supplies pairwise homographies and frame quality weights to robust scene consensus. Cross-modal QA produces a canonical product decision and a separate score-based ranking. Image panels use S02; the support grids and parameter-space cluster illustrate the computation. The editable schematic layout was revised with OpenAI Codex assistance.

4.2. Pairwise Evidence Generator

UAV-TAlign uses a pretrained multimodal matcher to generate pairwise evidence. The reference implementation adopts MINIMA with the RoMa branch, while the remaining stages accept homographies and geometry diagnostics from any compatible matcher [14, 16].

4.3. Progressive Evidence Scheduling

UAV-TAlign starts from a compact, evenly distributed frame subset and expands the support set in deterministic stages. The schedule scales with scene size and can cover the complete sequence. Expansion stops when accepted support, consensus stability, and cross-modal quality conditions are satisfied; otherwise, the next subset is evaluated. Scenes with at most 300 pairs use all available pairs. Larger scenes start with approximately 15% of the pairs and expand in 15-percentage-point increments up to 50%, followed by the full scene if needed. The 12-frame setting controls the representative QA probes, not a fixed cap on matching evidence. Exact support and stopping rules are specified in Appendix A.

Within each stage, evenly distributed sampling provides reproducible

temporal or index coverage. The random-selection variant is evaluated separately through a multi-seed sensitivity analysis.

4.4. Geometry-Informed Frame Weighting

For each selected pair, the matcher produces cross-modal correspondences and a frame-level homography estimate. UAV-TAlign measures match count, inlier support, inlier ratio, reprojection error, and spatial coverage. A frame enters the scene candidate pool when these statistics satisfy the geometry-quality criterion. This treatment follows established homography estimation and robust fitting practice [25, 24, 26, 27].

Each accepted frame also receives a continuous quality weight. The weight combines inlier ratio, spatial coverage, reprojection quality, matcher confidence, and inlier support, giving greater influence to homographies with stronger geometry. The fixed frame quality weight is

$$\begin{aligned} q_i = 0.42 r_i + 0.28 c_i + \frac{0.15}{1 + \max(e_i, 0)} \\ + 0.10 m_i + 0.05 \tanh(n_i/300), \end{aligned} \tag{2}$$

where r_i is inlier ratio, c_i is spatial match coverage, e_i is mean inlier reprojection error in RGB-image pixels, m_i is mean matcher confidence, and n_i is the number of inliers. The coefficients remain fixed across all UAV-TAlign-12K evaluations.

4.5. Robust Scene Consensus

Once enough frame-level homographies have been accepted, UAV-TAlign forms a robust scene consensus. Homographies are mapped into parameter space, centered with a median reference, and filtered by a median-absolute-deviation rule to suppress unstable estimates before weighted averaging. The retained homographies are then combined using the frame quality scores defined above. Let $\mathbf{h}_i$ contain the eight free entries of H_i after scaling its bottom-right entry to one. The consensus set is

$$\begin{aligned} \mathbf{h}_S^{\mathrm{med}} &= \mathrm{median}_{j \in \mathcal{A}_S}(\mathbf{h}_j), \\ d_i &= \left\| \mathbf{h}_i - \mathbf{h}_S^{\mathrm{med}} \right\|_2, \\ m_d &= \mathrm{median}_{j \in \mathcal{A}_S}(d_j), \\ s_d &= \mathrm{median}_{j \in \mathcal{A}_S} |d_j - m_d| + 10^{-9}, \\ \mathcal{C}_S &= \{ i \in \mathcal{A}_S : d_i \le m_d + 2.5 s_d \}. \end{aligned} \tag{3}$$

where $\mathcal{A}_S$ is the accepted-frame set before robust consensus. The scene transform is then obtained by quality-weighted aggregation,

$$\bar{\mathbf{h}}_S = \frac{\sum_{i \in \mathcal{C}_S} q_i \mathbf{h}_i}{\sum_{i \in \mathcal{C}_S} q_i}, \qquad \bar{H}_S = \mathrm{mat}(\bar{\mathbf{h}}_S). \tag{4}$$

The pipeline records stability diagnostics around the aggregate transform, including homography dispersion relative to the aggregate and the ratio of robustly rejected frame-level estimates. These diagnostics connect consensus formation with the final scene-product decision.

4.6. Cross-Modal Scene Verification

The final stage integrates geometric support with cross-modal scene consistency. UAV-TAlign evaluates the consensus transform on representative frames using edge-overlap improvement, gradient-NCC improvement, the fraction of improved frames, and severe relative outliers. It also records accepted-frame support, mean inlier ratio, mean reprojection error, spatial coverage, dispersion around the aggregate transform, and robust-rejection ratio.

The canonical decision combines accepted support with either a primary edge/gradient test or a baseline-aware path that additionally constrains geometry and outliers. A fixed, interpretable score summarizes normalized diagnostics for operating-profile visualization,

$$\begin{aligned} R(S) = 100\big(&0.25\hat{a}_S + 0.25\hat{o}_S + 0.20\hat{j}_S \\ &+ 0.15\hat{e}_S + 0.10\hat{g}_S + 0.05\hat{z}_S\big). \end{aligned} \tag{5}$$

Here each hatted component lies in $[0, 1]$, with larger values representing stronger accepted support, outlier control, consensus retention, edge improvement, gradient improvement, and geometric support, respectively. Appendix A.3 gives every normalization and clipping constant. The canonical decision requires a minimum number of accepted frame hypotheses and one of two fixed verification paths,

$$y_S = \mathbb{1}[N_S \geq N_{\min}]\,\mathbb{1}[L_S \vee W_S]\,. \tag{6}$$

Here N_S is the accepted-frame count and $N_{\min}$ is the scene-size-dependent support threshold. L_S denotes the primary edge/gradient path; W_S combines accepted support, consensus geometry, relative outlier control, and gradient

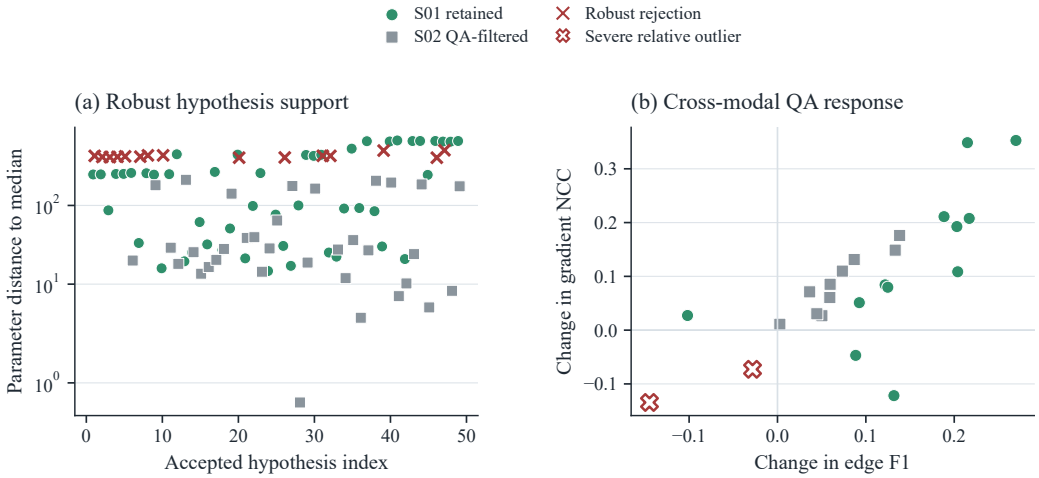

Figure 4: Robust consensus and cross-modal QA on the S01/S02 scene control. (a) Parameter-space distances identify hypotheses excluded by median/MAD filtering. (b) QA probe frames expose different edge/gradient responses and severe relative outliers. S01 and S02 have identical accepted support (49/50) but different canonical outcomes.

improvement when the initialization is size-based. Appendix A.2 specifies both paths. The score $R(S)$ orders scenes for reliability–coverage analysis, while y_S defines the canonical set of high-confidence alignment products.

Figure 4 exposes the internal evidence flow on a matched scene control: the same accepted-frame count can lead to different product decisions once consensus rejection and cross-modal outliers are incorporated.

5. Experimental Setup

5.1. Baseline Suite

We evaluate seven representative baselines without UAV-TAlign-12K-specific adaptation; all learned methods use public pretrained weights. The suite covers three method families. SIFT and AKAZE provide classical keypoint references, each followed by robust homography estimation. RIFT2 provides an infrared-visible-specific classical reference through its Python implementation [29, 51]. The modern suite contains Kornia LoFTR, RoMa outdoor, XoFTR-640, and raw MINIMA [10, 14, 15, 16].

Raw MINIMA uses the `minima_roma` checkpoint family with the full RoMa branch. RoMa uses the public outdoor full model, and XoFTR uses its 640-resolution visible-thermal checkpoint. LoFTR uses Kornia's public

outdoor-pretrained model. Unified inference wrappers standardize image I/O, model output, homography fitting, and result schemas while preserving each pretrained model.

5.2. Geometric Fitting and Deployment Profiles

All pairwise methods use USAC-MAGSAC for homography estimation. The shared profile uses a 3.0-pixel reprojection threshold, 0.999 confidence, and 10,000 maximum iterations; RIFT2 retains its separately specified 5.0-pixel profile. Spatial match coverage is measured on the same 4×4 grid for all methods. Appendix A.4 defines the fitting diagnostics and their aggregation.

RIFT2 uses its frozen resize-aware profile: `max_dim=1200`, `npt=2000`, `patch_size=64`, Lowe ratio 0.95, and USAC-MAGSAC with a 5.0-pixel reprojection threshold. LoFTR uses resize-aware inference with a longest image dimension of 1200 and automatic mixed precision. Runtime and fitting diagnostics are reported for these declared deployment profiles.

5.3. Protocol Validation

Two controlled experiments evaluate the scene protocol on frozen formal outputs. Five-fold leave-frame-out analysis partitions accepted frame hypotheses deterministically, withholds one fold, recomputes scene consensus, and measures grid displacement from the full-evidence transform. A controlled sensitivity test applies translation, rotation, and scale perturbations at nominal severities of 2, 5, 10, 20, and 40 pixels to three representative frames per scene. The test records edge-overlap and gradient-NCC responses while keeping the canonical decision fixed.

Journal-scale experiments were conducted in isolated Python environments on an NVIDIA RTX 4090 workstation. The main environment used Python 3.10, PyTorch 2.0.1, torchvision 0.15.2, CUDA 11.8, OpenCV 4.11.0, and Kornia 0.6.11; RIFT2 used an independent Python 3.11 environment. Every run used the official manifest, fixed model weights or interfaces, isolated output roots, and read-only raw dataset directories.

Implementation and visualization assistance.. OpenAI Codex assisted with Python implementation, evaluation wrappers, result aggregation, and plotting scripts. The reported quantities are computed by the versioned scripts from the fixed manifest and archived run outputs. Image panels use captured UAV images, frozen homographies, and deterministic cropping, warping, and edge-overlay operations; no generative image model creates or modifies the captured scene content.

6. Results

6.1. Pairwise Infrared-Visible Homography Evidence

Pairwise infrared-visible homography evidence is broadly available on UAV-TAlign-12K. Table 2 reports the full 6,037-pair official evaluation split. SIFT and AKAZE establish classical keypoint references with 87.08% and 94.29% homography availability, respectively. RIFT2, raw MINIMA, and RoMa produce homographies for every evaluated pair, while LoFTR and XoFTR exceed 99% availability. These results establish infrared-specific and modern dense matchers as strong evidence generators for scene-scale alignment.

Table 2: Pairwise infrared-visible homography evidence on UAV-TAlign-12K. IR and Cov. are mean inlier ratio and mean candidate-match coverage. Reproj. is the median of pair-level mean inlier residuals (pixels); Time is mean seconds per pair. Statistics use recorded finite values (Appendix A.4). These are complementary diagnostics, not a single quality ranking.

Method	H / N	H (%)	IR	Cov.	Reproj.	Time
Classical keypoint						
SIFT	5257/6037	87.08	0.242	0.627	0.000	4.421
AKAZE	5692/6037	94.29	0.160	0.730	0.105	1.103
Infrared-visible classical						
RIFT2 (Python, resize-aware)	6037/6037	**100.00**	0.086	0.354	0.121	13.980
Modern dense / learned						
Kornia LoFTR	5984/6037	99.12	0.074	0.895	0.823	0.169
XoFTR-640	5989/6037	99.20	0.089	0.949	1.732	0.647
RoMa outdoor	6037/6037	**100.00**	0.328	0.997	1.741	0.929
raw MINIMA	6037/6037	**100.00**	0.319	1.000	1.773	1.346

The method families exhibit complementary evidence profiles. RIFT2 combines complete homography availability with low fitted reprojection error, whereas raw MINIMA and RoMa provide nearly full spatial coverage. Jointly, availability, support, and coverage provide the geometric context for the scene-level benchmark track.

6.2. Scene-Level Reference Result

UAV-TAlign converts the pairwise evidence into scene transforms for all 15 scenes. At the canonical operating point, it identifies nine high-confidence

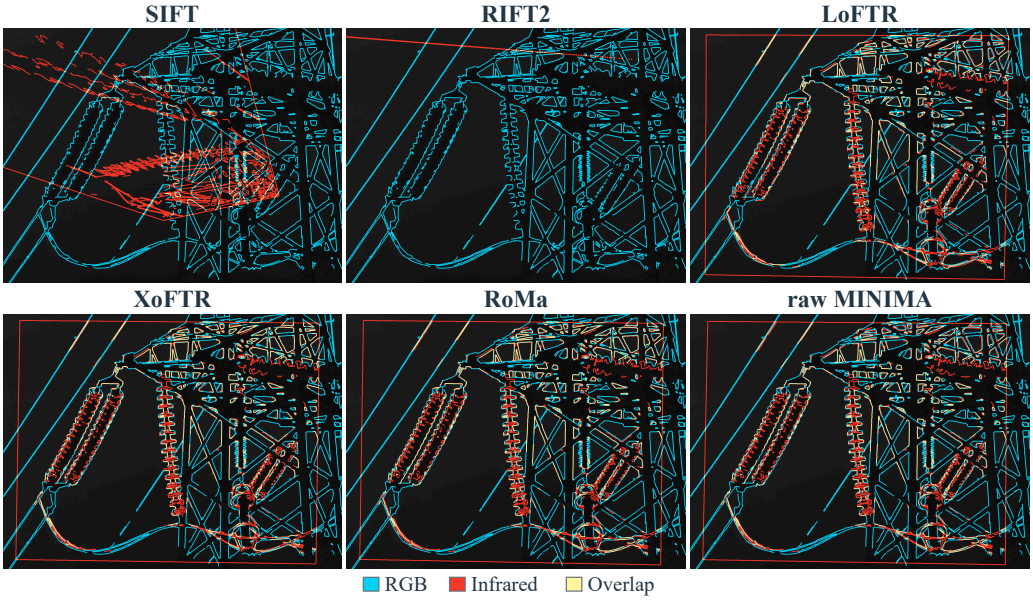

Figure 5: Fixed-pair homography evidence from six representative matchers on S04/000020. All panels use the same RGB coordinate system and edge-display settings: cyan denotes RGB edges, red denotes warped infrared edges, and pale yellow denotes overlap. Transforms follow the declared fitting profiles. Full-split availability and coverage are reported in Table 2.

products, representing 60.0% scene coverage and 74.4% pair coverage. The accepted-frame support within these products reaches 1491/2304 (64.7%). Across all 15 scenes, accepted support is 2473/3847 (64.3%); the two ratios use different scene sets.

Table 3: Canonical scene-level operating point for UAV-TAlign on UAV-TAlign-12K. Pair coverage and accepted-frame support are computed over the nine high-confidence scene products.

UAV-TAlign scene-level measure	Value
Homographies available	15/15
Canonical retained scenes	9/15
Scene coverage	60.0%
Pair coverage	74.4%
Retained-set accepted / attempted	1491/2304
Retained-set accepted ratio	64.7%

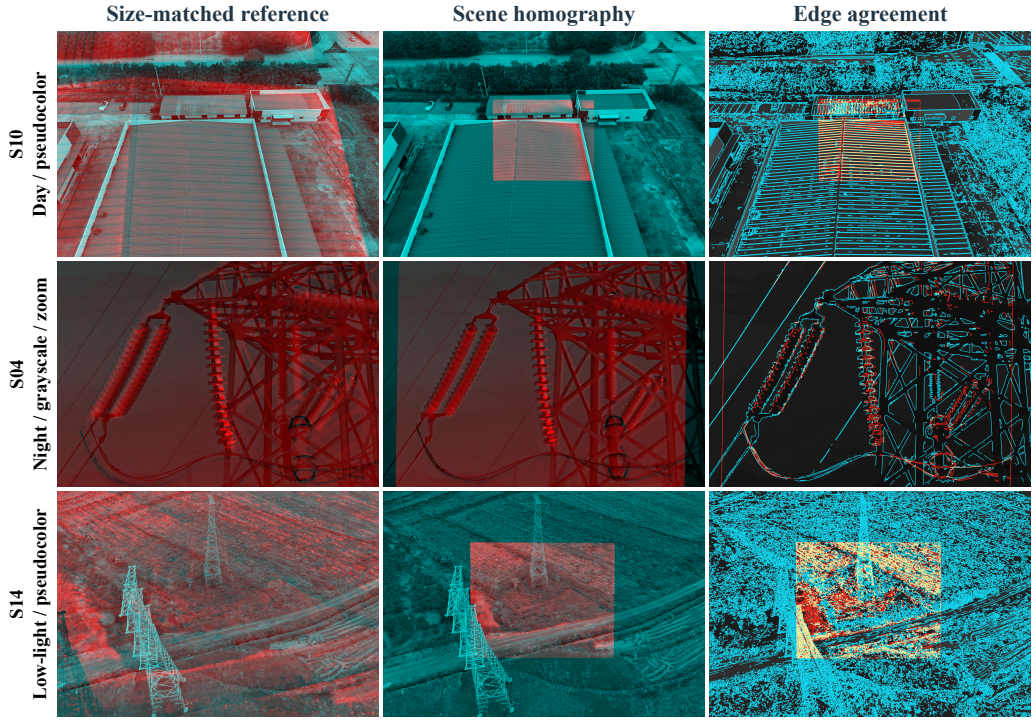

Figure 6: Representative high-confidence UAV-TAlign scene products at the canonical operating point. Rows cover day, night, and low-light scenes with grayscale and native pseudocolor infrared rendering. Columns show the size-matched reference, scene-homography overlay, and edge agreement, using a fixed crop within each row. RGB and infrared inputs are shown in Figure 1; an additional tower example appears in Appendix B.

Figure 6 presents representative canonical alignment products. They span illumination, rendering, and view regimes, while shared crops and edge-agreement views make the geometric improvement directly inspectable across modalities.

6.3. Condition-Aware Operating Profiles

Illumination produces distinct scene-product and accepted-support profiles. Day capture provides the highest scene coverage and accepted-frame ratio. Night capture yields 3/5 high-confidence products, while low-light capture preserves a 61.3% accepted-frame ratio across its three scenes. These complementary measurements separate available frame support from the final scene product.

Table 4: Condition-aware operating profile by illumination. Accepted/attempted aggregates frame support over all scenes in each condition.

Condition	Retained / scenes	Scene coverage (%)	Accepted / attempted	Accepted ratio (%)
Day	5/7	71.4	1650/2203	74.9
Night	3/5	60.0	361/890	40.6
Low-light	1/3	33.3	462/754	61.3

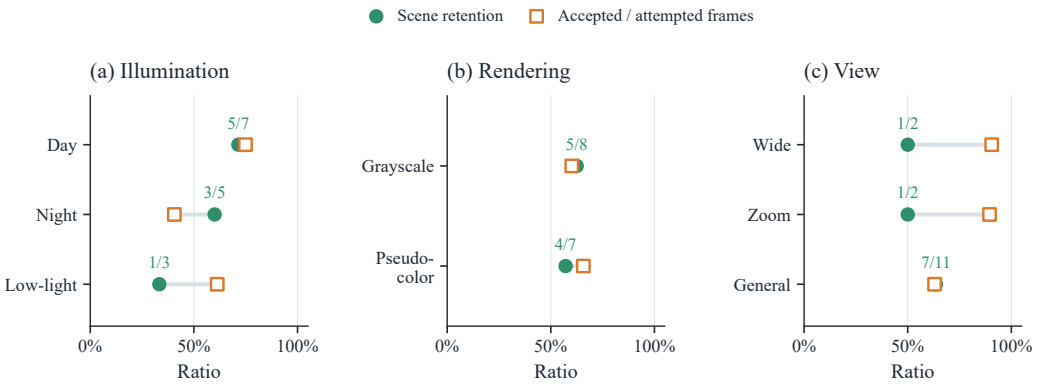

Figure 7: Condition-aware operating profiles by (a) illumination, (b) infrared rendering, and (c) view type. Filled circles show scene retention; open squares show accepted/attempted frame support. Labels give retained/total scenes. The same scenes recur across attribute groups.

Figure 7 extends this analysis to illumination, infrared rendering, and view type, reporting scene retention together with accepted-frame support.

6.4. Reliability-Controlled Scene Selection

Matched scene controls reveal why pairwise support alone does not determine the final product set. Figure 8 compares the day wide/zoom control under identical display conventions. The night control and all-scene QA plot are presented separately in Appendix B. Canonical products concentrate in the low-outlier region, whereas accepted-frame ratio spans both retained and QA-filtered scenes.

6.5. Reliability-Coverage Operating Profile

The fixed cross-modal verification rule defines the canonical product set. The interpretable score orders scenes into discrete reliability–coverage oper-

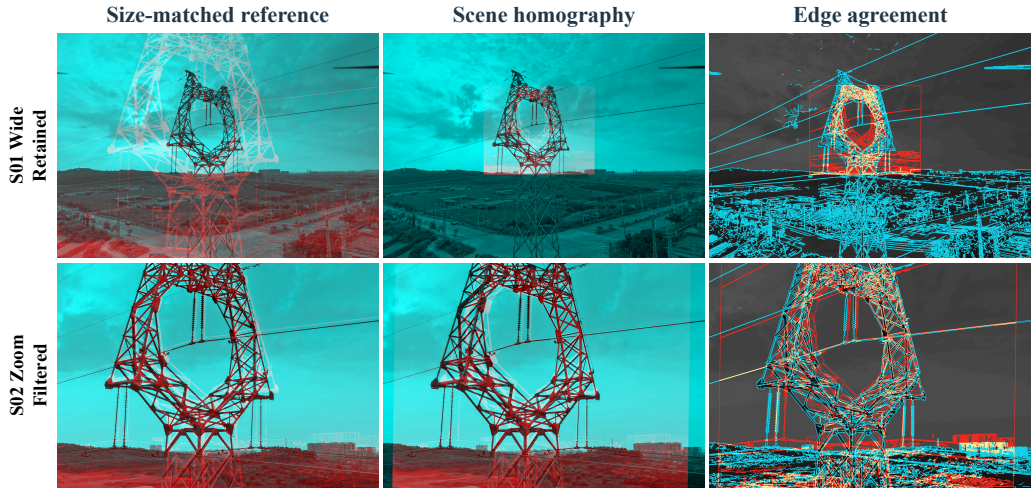

Figure 8: Day wide/zoom control: S01 is canonically retained, while S02 is QA-filtered. Columns show the size-matched reference, estimated scene homography, and edge agreement. The S02 estimate is displayed for inspection, not as a retained product. Canonical labels describe scene-level decisions, rather than separate decisions on the displayed frames.

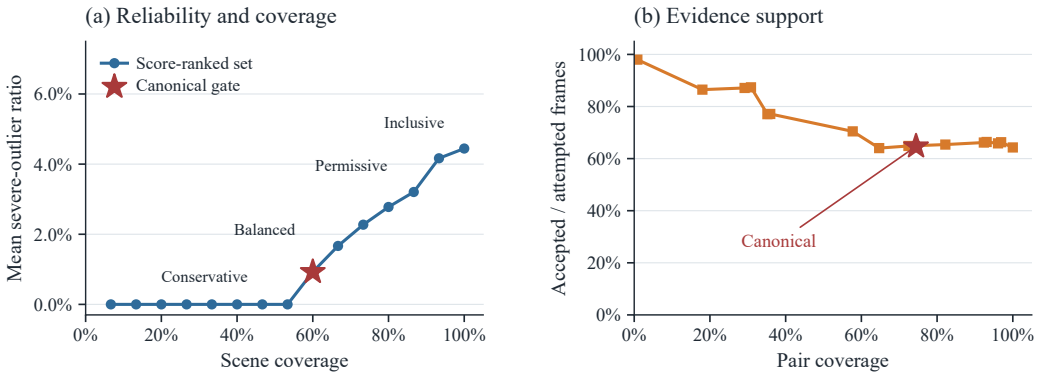

Figure 9: Reliability–coverage operating profile on UAV-TAlign-12K. (a) The score-ranked curve traces mean severe-outlier ratio as scene coverage increases; the star marks the independent canonical QA point, not the score-ranked top-nine set. (b) Pair coverage and accepted-frame support expose the corresponding evidence footprint. Lines connect observed operating points.

ating points. Figure 9 plots this score-ranked profile and overlays the fixed canonical operating point.

At the canonical point, UAV-TAlign identifies nine high-confidence products, covers 60.0% of scenes and 74.4% of evaluated pairs, and records a mean

severe-outlier ratio of 0.009. The score-ranked points expose the diagnostic trade-off as additional scenes enter the selected set.

6.6. Controlled Validation of Scene Diagnostics

The scene diagnostics respond consistently under independent resampling and controlled geometric perturbations. Five-fold leave-frame-out analysis covers all 15 scenes and 75 consensus recomputations. With approximately one fifth of accepted frame hypotheses withheld per fold, the median of the 15 scene-wise mean displacements from the full-evidence consensus is 1.39 pixels; the mean of those scene means is 2.26 pixels.

The perturbation test covers 45 representative frames and 675 controlled trials. Within each perturbation family, increasing severity produces a monotonic decrease in the mean joint edge/gradient quality signal. At the largest severity, translation, rotation, and scale produce mean joint-score changes of -0.074, -0.031, and -0.025, respectively. The corresponding degradation rates are 84.4%, 80.0%, and 73.3%. Table 5 summarizes the independent evidence. Figure 10 visualizes the full scene-wise resampling response and the severity-dependent perturbation trends.

Table 5: Controlled validation of scene-consensus stability and cross-modal quality sensitivity. Synthetic degradation rates and joint-score changes are reported at the largest tested severity.

Validation	Scope	Primary response	Degraded trials (%)
Leave-frame-out consensus	15 scenes / 75 folds	1.39 px median of scene means	–
Synthetic translation	45 frames / 225 trials	ΔJoint $= -0.074$	84.4
Synthetic rotation	45 frames / 225 trials	ΔJoint $= -0.031$	80.0
Synthetic scale	45 frames / 225 trials	ΔJoint $= -0.025$	73.3

6.7. Cumulative Ablation

The fixed 8-scene ablation isolates robust consensus and cross-modal verification. Robust consensus raises mean edge gain from 0.124 to 0.146 and gradient-NCC gain from 0.126 to 0.149. It simultaneously reduces the severe-outlier ratio from 0.104 to 0.042 and transform dispersion from 151.2 to 52.5

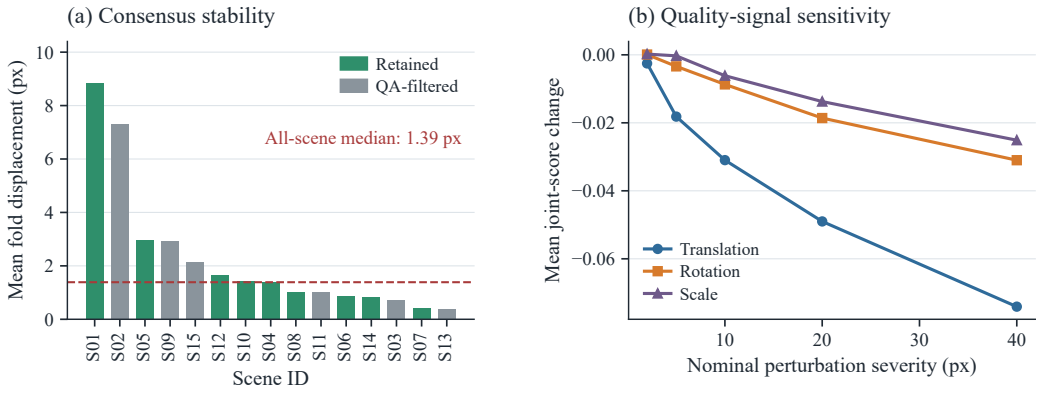

Figure 10: Controlled validation on UAV-TAlign-12K. (a) Mean five-fold leave-frame-out displacement for each scene; the dashed line marks the median of scene means. (b) Mean joint quality change over 45 frames per perturbation mode and severity.

pixels. Cross-modal verification then maps the stage output to the canonical product decision.

Table 6: Cumulative ablation on the fixed 8-scene UAV-TAlign-1K-Lite subset. Output/N counts available transforms in the first two rows and canonically retained scenes in the last row. Diagnostic means use all eight scenes at each stage; even selection is fixed throughout.

Variant	Con-sensus	QA	Output /N	ΔEdge ↑	ΔGrad ↑	Severe ↓	Disp. (px) ↓
Pairwise evidence only	–	–	8/8	0.124	0.126	0.104	151.2
+ Robust scene consensus	✓	–	8/8	0.146	0.149	0.042	52.5
+ Cross-modal verification	✓	✓	5/8	0.146	0.148	0.042	57.7

Figure 11 connects the cumulative module effects with the multi-seed selection study: robust consensus supplies the largest diagnostic gain, while the deterministic schedule provides a stable reference operating policy.

Across random-selection seeds, the same fixed subset yields 5/8 to 7/8 scene products while mean accepted-frame support remains comparatively stable. Deterministic-even selection provides a reproducible reference, while the random runs localize seed sensitivity to a small set of near-threshold scenes.

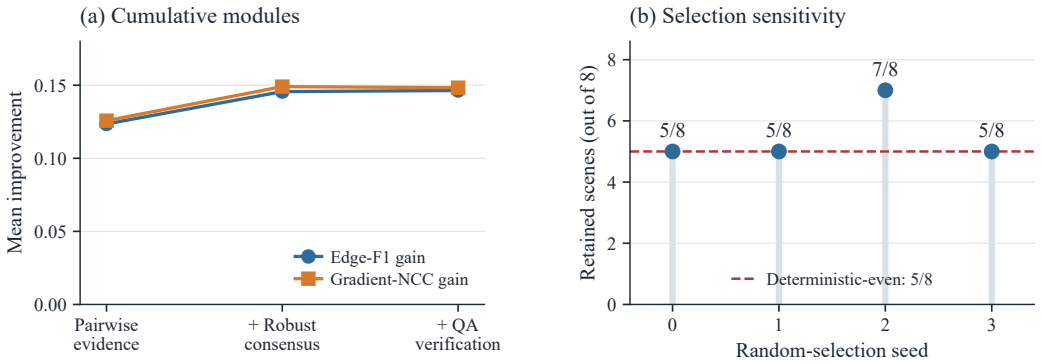

Figure 11: Module attribution on the fixed eight-scene UAV-TAlign-1K-Lite subset. (a) Cumulative changes in edge and gradient agreement. (b) Canonical retained-scene counts for four random seeds; the dashed line denotes the deterministic-even reference.

7. Discussion

7.1. From Pairwise Evidence to Scene-Scale Geometry

UAV-TAlign-12K reveals a transition in the UAV infrared-visible alignment problem. Infrared-specific and modern matchers generate homographies for nearly every pair, providing abundant geometric evidence across the official split. The evidence nevertheless varies in spatial support, cross-modal agreement, and consistency with other frames from the same scene. Scene consensus is therefore a distinct alignment stage: it converts heterogeneous frame estimates into one scene transform that can be used as a common geometric product.

The ablation results identify robust consensus as the dominant module-level contribution. Relative to pairwise evidence alone, consensus improves edge and gradient agreement while sharply reducing severe-outlier incidence and transform dispersion. Cross-modal verification then maps this stable geometry to the canonical high-confidence product set. The combined design links local correspondence evidence with the scene-scale output required by UAV inspection and multimodal sensing.

7.2. Coverage-Aware Operating Analysis

The reliability–coverage profile adds an operating dimension to fixed-threshold evaluation. The canonical point provides nine high-confidence products covering 74.4% of evaluated pairs, while the score-ranked sweep

exposes broader coverage regimes. This representation allows an application to select scene-product coverage according to its quality requirements without changing the pairwise matcher or consensus estimator.

The controlled analyses probe the underlying diagnostics. With one fold withheld, recomputed consensus yields a 1.39-pixel median of scene-wise mean displacements, characterizing sensitivity to the selected evidence. Translation, rotation, and scale perturbations progressively reduce the mean joint edge/gradient signal, characterizing its response to geometric disturbance. These observations support the use of consensus stability and image agreement as complementary diagnostics.

7.3. Benchmark Extensibility

The benchmark separates pairwise correspondence, scene aggregation, and cross-modal verification through a manifest-based interface. Future matchers and consensus estimators can therefore be studied on the same captured-data split. The scene homography describes a dominant planar or far-field alignment; spatially varying geometry calls for local or nonrigid representations. The reported operating profiles characterize relative image agreement and evidence consistency within the 15 captured scenes.

Deterministic evidence scheduling provides a stable default for this shared interface. The multi-seed study shows that random selection changes the product status of a small set of near-threshold scenes, whereas accepted-frame support remains comparatively stable. This result positions scheduling as a reproducibility mechanism and leaves robust consensus and cross-modal verification as the principal scene-product components.

8. Conclusion

UAV-TAlign-12K connects captured UAV infrared-visible imagery with evaluation of pairwise evidence, scene consensus, and operating coverage. Across its 6,037-pair official split, seven baselines show that infrared-specific and modern matchers provide abundant homography evidence. The remaining task is to consolidate that evidence and decide which scene transforms to retain. UAV-TAlign produces transforms for all 15 scenes and retains nine, covering 74.4% of evaluated pairs. Cumulative ablations identify robust consensus as the main source of improved image agreement and reduced dispersion; resampling and controlled perturbations characterize stability and diagnostic sensitivity. Together, the dataset and reference framework make

scene-scale evidence consolidation an explicit, reproducible evaluation target for UAV infrared-visible alignment.

Data Availability

UAV-TAlign-12K is jointly owned by Nanning Normal University and Guangxi University and is prepared for release under the Creative Commons Attribution 4.0 International license. The official evaluation split contains 6,037 integrity-checked RGB-infrared pairs from the 6,039-pair collection. The image archive awaits repository deposit and assignment of an access link. A separate reproducibility package contains the official manifest, metadata, integrity summaries, evaluation code, and numerical figure sources for submission as supplementary material. Original UAV-TAlign software is licensed under Apache-2.0; third-party components retain their respective licenses.

Appendix A. Protocol Definitions and Reproducibility Details

Appendix A.1. Probe Frames and Relative Quality Signals

Each scene uses up to 12 evenly distributed manifest pairs as QA probes, selected from the scene rather than only from accepted matches. Images are evaluated at quarter scale with the longest dimension capped at 640 pixels. The scene homography is converted to these evaluation coordinates. For each probe, both the transformed image and the size-matched reference are scored within the same warped infrared validity mask. The reference resizes infrared edges and gradients to the RGB evaluation grid. Its full-image size mapping is denoted by H_0; it is not a fitted registration.

Let $E_i(H)$ denote edge-overlap F1 with a one-pixel dilation tolerance, and $G_i(H)$ denote normalized cross-correlation between gradient magnitudes inside the validity mask. Define

$$\delta E_i = E_i(\bar{H}_S) - E_i(H_0), \qquad \delta G_i = G_i(\bar{H}_S) - G_i(H_0). \tag{A.1}$$

The scene gains ΔE_S and ΔG_S are probe means. A severe relative outlier has both $\delta E_i < -0.01$ and $\delta G_i < -0.01$; O_S is its fraction among probes. Let $p_\vee$ be the fraction with either gain positive and p_G the fraction with positive gradient gain. These image-domain diagnostics measure agreement relative to the size-matched reference.

Appendix A.2. Support and Canonical Verification

A frame enters the accepted pool when homography fitting succeeds with at least 80 candidate matches, inlier ratio at least 0.30, mean inlier reprojection error at most 4 pixels, and coverage at least 0.25. For a scene with N valid pairs, the frozen minimum accepted support is

$$N_{\min} = \max(4, \min(N, \max(8, \min(20, \mathrm{round}(0.15N))))) \,. \tag{A.2}$$

This yields thresholds of 8 for S01–S04, 15 for S07, and 20 for the remaining scenes. Accepted support A_S is the number of accepted frames divided by the number attempted. The robust-rejection ratio J_S is the fraction removed from this accepted pool during consensus. Dispersion D_S is the median, across accepted hypotheses, of mean displacement from the consensus on a 5×5 infrared-image grid, measured in RGB-image pixels.

The primary path L_S requires $\Delta E_S > 0$, $\Delta G_S > 0$, $p_\vee \geq 0.60$, and zero severe relative outliers. The alternative path W_S applies to infrared input with size-based initialization and requires all of the following:

$$\begin{aligned} &A_S \geq 0.80, && \bar{r}_S \geq 0.30, && \bar{e}_S \leq 4 \text{ px}, && \bar{c}_S \geq 0.25, \\ &D_S \leq 25 \text{ px}, && J_S \leq 0.25, && O_S \leq 0.10, && n_{\text{out}} \leq 1, \\ &\Delta E_S \geq -0.05, && \Delta G_S > 0, && p_G \geq 0.60. \end{aligned} \tag{A.3}$$

Here $\bar{r}_S, \bar{e}_S, \bar{c}_S$ are accepted-frame means. Path L_S gives a pass; path W_S gives a pass with warning when L_S is not met. Both contribute to canonical retention once $N_S \geq N_{\min}$. Adaptive expansion stops when this retention condition and the $D_S \leq 25$, $J_S \leq 0.25$ stability checks hold. After exhausting the schedule, the final canonical decision still follows Eq. 6; the stopping checks are not an additional requirement on the primary path. Thus early stopping and final retention have distinct roles.

Appendix A.3. Normalized Reliability Score

Let $C(x) = \min(1, \max(0, x))$. The components used in Eq. 5 are

$$\begin{aligned} &\hat{a}_S = C(A_S), && \hat{o}_S = 1 - C(O_S/0.10), \\ &\hat{j}_S = 1 - C(J_S/0.25), && \hat{e}_S = C((\Delta E_S + 0.05)/0.20), \\ &\hat{g}_S = C(\Delta G_S/0.15), \end{aligned} \tag{A.4}$$

$$\hat{z}_S = 0.5C(\bar{r}_S/0.50) + 0.3C(\bar{c}_S) + 0.2\,[1 - C((\bar{e}_S - 1.5)/1.0)] \,. \tag{A.5}$$

All constants are fixed and no model is fitted to scene outcomes. The resulting $[0, 100]$ score provides a diagnostic ordering, not a calibrated probability. Score-ranked coverage points select the first K scenes in this ordering. The canonical point is evaluated on its own retained set.

Appendix A.4. Metric Aggregation

Pairwise statistics.. Inlier ratio is the number of fitted inliers divided by the candidate-match count. Coverage is the fraction of the 16 infrared-image grid cells occupied by candidate matches, before robust fitting. Reprojection error is the mean Euclidean inlier residual in RGB-image pixels for one pair; the main table reports the median of these pair-level means. Each summary uses the recorded finite, non-null values of that metric. Recorded zero inlier ratios and coverages are included, whereas missing reprojection values are excluded. Runtime is the mean recorded time per pair under the stated implementation and deployment profile.

Scene and condition statistics.. Scene coverage is the fraction of scenes retained. Pair coverage weights each retained scene by its valid manifest pair count. Accepted/attempted ratios use the summed frame counts within the indicated scene set. Accordingly, retained-set support is $1491/2304 = 64.7\%$, whereas all-scene support is $2473/3847 = 64.3\%$. Condition retention is scene-macro; accepted support is frame-micro.

Resampling and perturbations.. For leave-frame-out analysis, grid displacement is averaged within each fold, then across the five folds of each scene. The reported 1.39 pixels is the median of these 15 scene means; their mean is 2.26 pixels. Perturbation curves average the joint edge/gradient score change over 45 frames at each severity and mode. A degradation rate counts frames with a negative joint score change at that setting. These tests assess consensus stability and quality-signal response, respectively, rather than fitting or recalibrating the ranking score.

Appendix B. Dataset Distribution and Additional Visual Evidence

Figure B.1 reports per-scene counts and acquisition attributes. Figures B.2 and B.3 extend the main-text examples to a tower-orbit sequence and the night zoom/wide control. All displays use frozen estimates and the same image crops as the corresponding source records.

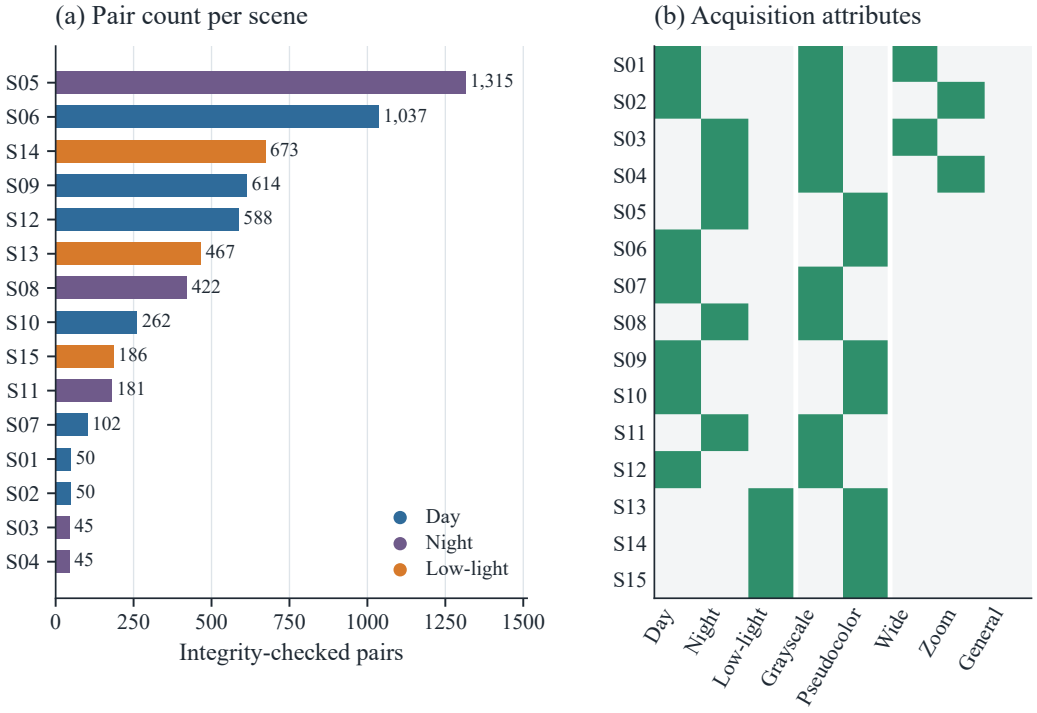

Figure B.1: Official evaluation split: (a) valid pair counts and (b) acquisition attributes of the 15 scenes. Color in (a) identifies illumination; filled cells in (b) identify scene attributes.

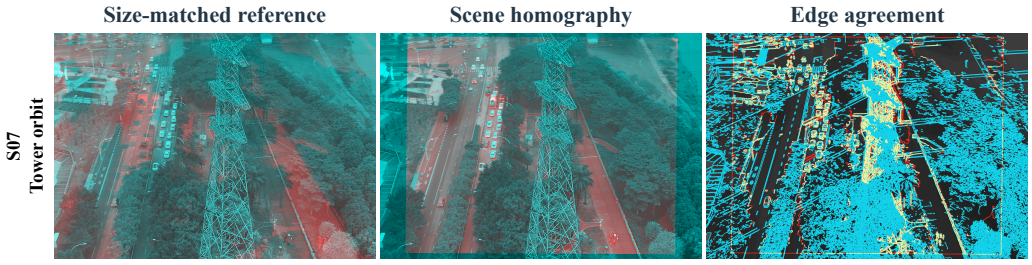

Figure B.2: Additional canonical product from the S07 tower-orbit sequence. The panels compare the size-matched reference, scene-homography overlay, and edge agreement on the same display crop.

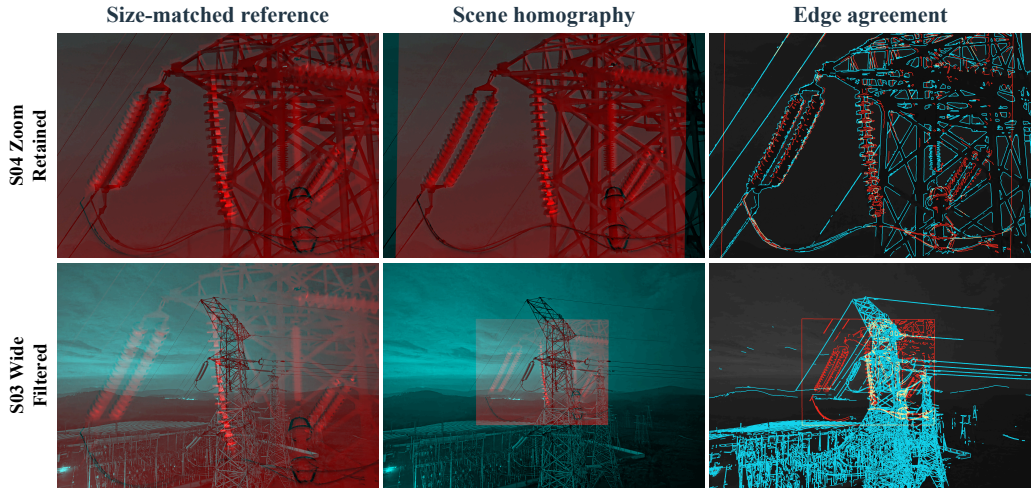

Figure B.3: Night zoom/wide control. S04 is canonically retained and S03 is QA-filtered. Each row shows the size-matched reference, estimated scene homography, and edge agreement. The filtered estimate remains visible for inspection.

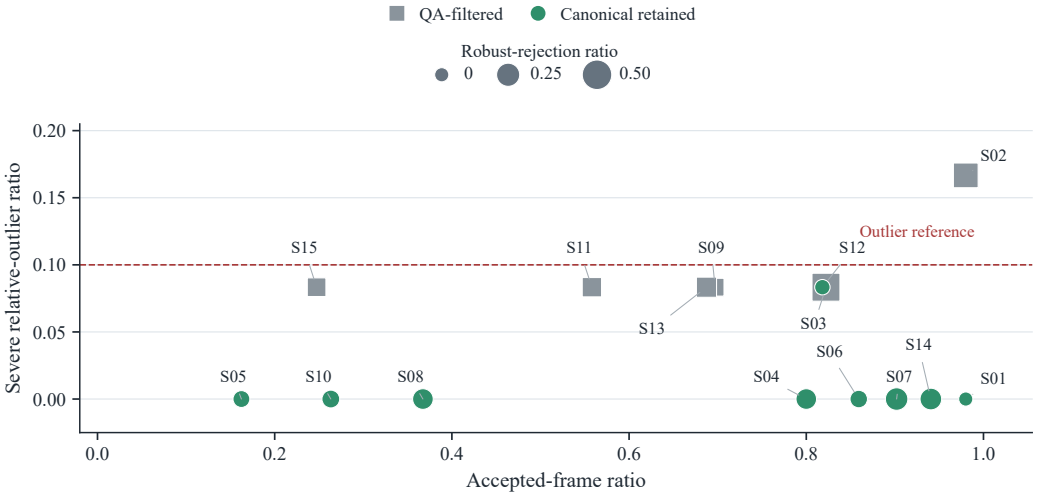

Figure B.4: All-scene QA diagnostics. Circles denote canonical retained scenes and squares denote QA-filtered scenes; marker size represents robust-rejection ratio. The dashed line marks the 0.10 severe-outlier-ratio reference used in the baseline-aware path. The full canonical decision combines the verification paths and support requirement in Appendix A.2.

Declaration of Generative AI and AI-Assisted Technologies

OpenAI ChatGPT and Codex assisted with language revision, translation, manuscript organization, LaTeX preparation, and editable schematic layout. AI-assisted code development and data visualization are described in the experimental setup. The authors retain responsibility for reviewing and verifying the content and for the final manuscript.

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